

ICT Classified -Chapter (6) - ICT Applications

1

Expert systems have many applications.

(a) Name two applications that use expert systems.

1.....

2.....

[2]

(b) Tick three components of an expert system.

	Tick (✓)
Interactive user interface	
Spreadsheet	
Motor	
Printer	
Knowledge base	
Search engine	
Actuator	
Rules base	

[3]

Q 9 - Page 12 0417/12 May/June 2019

A train company is improving security on its journeys. They are planning to introduce a facial recognition system for passengers using their trains.

[illegible]

Tick (✓) whether the following are the most appropriate examples of biometrics or contactless or physical direct data entry.

	biometrics (✓)	contactless (✓)	physical (✓)
RFID			
Magnetic stripe			
Retina scan			
Voice recognition			

3

The worker has set the heater control to switch on at 16:00 for two hours.

Explain how the microprocessor can operate the automatic heater to turn on and off at the correct times.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[6]

Q 10 - Page 14 – b 0417/13 May/June 2020

4

Smartwatches can be used in remote regions of the world. These can have an in-built GPS system to help the wearer to locate their position and therefore help with navigation.

(a) Explain how GPS is used to display their location on the watch.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

(b) Describe the advantages of using a navigation system with GPS rather than printed maps in this scenario.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

Q 7 - Page 5 0417/11 May/June 2018

5 The school is planning to use fingerprints as a way to record when students arrive in the class.

Describe the advantages and disadvantages of using fingerprint data entry for student registration systems rather than magnetic ID cards.

.....

.....

.....

.....

.....

.....

.....

.....

.....

[4]

Q 10 - Page 13-b 0417/11 May/June 2020

6

Many car parks are now fitted with CCTV that can read car number plates.

(a) Explain how number plate recognition systems work.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

(b) Describe **two** problems that could occur when the system reads a number plate .

1.....

.....

2.....

.....

[2]

Q 13 - Page 15 0417/11 May/June 2018

Q 8 - Page 11 0417/12 May/June 2019

9

The doctors in the health centre use an expert system to help them diagnose illnesses.

The symptoms of the patient's illness are typed in by the doctor.

Describe the processing that the system uses so that it can suggest possible illnesses.

This image shows a full page of white paper with ten horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and extend across the entire width of the page. There is no text or other markings on the paper.

[5]

Q 8 - Page 9 - e 0417/13 May/June 2019

A large city centre hotel uses a computer-controlled barrier for entry into its car park. A light sensor is used to detect the presence of a car and prevent the barrier from dropping onto the car.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

[6]

Q 9 - Page 10 0417/13 May/June 2019

11

Expert systems are used by doctors.

(a) Describe how an expert system can be used to diagnose illnesses.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[5]

(b) Name two other applications of expert systems.

1

2

[2]

Q 11 - Page 10 0417/11 May/June 2018

12

A farmer has purchased a computerised feeding system for her goats. A goat has an RFID tag attached to its ear, which is recognised by the computer. The system uses a passive RFID tag.

(a) Describe how the RFID tag can be activated.

.....

.....

.....

.....

.....

.....

[3]

(b) The system recognises the goat and therefore gives the correct feed to the animal.

Describe how RFID technology will be used to give the correct feed to the animal.

.....

.....

.....

.....

.....

.....

[3]

Q 7 - Page 5 0417/12 May/June 2018

13	Automatic car parking systems built into cars are becoming more common. They allow the car to park safely in a parking bay unaided. The car uses proximity sensors fitted to the front and rear of the car to help it park. There are many advantages to using these sensors. (a) State three disadvantages of using proximity sensors in this way. 1 2 3 [3]
	(b) Modern-day drivers rely less on paper maps and more on satellite navigation systems (satnav). Describe the inputs, outputs and processing of a satnav when used by a driver to reach a destination. [4] Q 9 - Page 10 0417/12 May/June 2018

The European Space Agency (ESA) is building a new space telescope to orbit the Earth and search for distant galaxies. The ESA is using computer controlled robots to build the lens of the telescope.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

[6]

Q 9 - Page 10 0417/12 May/June 2018

15

Adam is paying his bill in a restaurant using a contactless debit card.

Describe the computer processing involved in Electronic Funds Transfer (EFT) using contactless cards.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[4]

Q 15 - Page 15 b 0417/12 May/June 2018

16

A large farm uses a robotic system to milk and feed animals. Until recently this would have been carried out by farm workers.

Discuss the advantages and disadvantages of using robotics in automated farms.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[6]

Q 8 - Page 7 0417/13 May/June 2018

17

An automatic washing machine contains a device to read RFID chips sewn into clothing. When an item of clothing is placed in the washing machine it is scanned.

(a) Describe how the data is read from the RFID chip.

.....

.....

.....

..... [2]

(b) Explain the benefits of using RFID technology in a washing machine.

.....

.....

.....

..... [2]

Q 9 - Page 5 0417/11 May/June 2017

18

The city of Venice is in danger of being damaged due to rising sea levels. A new computer controlled flood defence system is being planned. The developers of the system are using a spreadsheet to model the plan.

(a) Give **three** reasons why computer models are used to simulate the flood defence system.

- 1.....
.....
- 2.....
.....
- 3.....
.....

[3]

(b) The city uses a manual system for detecting flood waters. This relies on people watching for the rising levels of water.

Explain why creating a computerised system is better than using the manual system.

-

[4]

19

- (c) The developers have used the results from the model to choose a computer controlled flood defence system. It will have a barrier which can open or close.

Describe the role of the microprocessor in such a system.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[5]

Q 10 - Page 6 0417/11 May/June 2017

20

Robots are being used in the mining industry to dig up minerals from under the surface of the Earth.

(a) Give **three advantages** of using robots rather than humans to do this task.

1.....

.....

2.....

.....

3.....

.....

[3]

(b) Give **two disadvantages** of using robots rather than humans.

1.....

.....

2.....

.....

[2]

Q 10 - Page 6 0417/11 May/June 2017

21

The managers of a nuclear power plant need to check the radioactivity levels in a chimney stack which needs to be demolished. They intend to use a robot. This device has sensors to read the radioactivity levels.

- (a) Explain why the data from the sensors cannot be read directly by the microprocessor and name the device needed to change the data into a form that the microprocessor can understand.

.....

.....

.....

.....

.....

..... [3]

- (b) The managers have been told they can use this robot for other tasks by replacing the existing sensors with different ones.

Give **three** advantages of using an existing robot device rather than developing a new one.

.....

.....

.....

.....

.....

..... [3]

Q 5 - Page 4 0417/12 May/June 2017

22

Driverless automobiles use a number of different methods to drive between two points.

(a) Name a sensor that could be used on the automobile to detect a person walking in front of it.

..... [1]

(b) Modern automobiles contain electronic devices designed to help the driver.

Suggest **three** examples of labour-saving or safety devices in a modern automobile that use sensors.

1.....

.....

2.....

.....

3.....

.....

[3]

(c) Discuss the advantages and disadvantages of these devices in relation to automobile journeys.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

Q 7 - Page 5 0417/12 May/June 2017

23

Expert systems are used in medical applications.

(a) From the list below, tick **two** components of an expert system.

	tick (✓)
Control unit	
Graphics program	
Inference engine	
RFID	
Rules base	
Search engine	

[2]

(b) Describe **three** benefits of the doctor using an expert system to help diagnose a patient's illness.

.....

.....

.....

.....

.....

.....

.....

.....

[3]

People can manage their bank accounts in different ways. One way is telephone banking.

- Describe the steps needed to do this.

[illegible]

[6]

- List **four** items of data found on a debit card which can be read by computer.

- 1.....
- 2.....
- 3.....
- 4.....

25	Key-loggers are thought to be security risks but they can be used to monitor and track the use of employees on a computer system. (a) Describe how a key-logger monitors computer usage. [2] (b) A small minority of employees in a company have been using the computers for non-business use, so the company has had to introduce other methods of monitoring their use of ICT. Apart from key-logging list three other ways that employees could be monitored in their use of ICT. 1 2 3 [3] Q 13 - Page 12 0417/13 May/June 2017
26	Describe three benefits to customers of using internet banking. 1 2 3 Q 9 - Page 5 - b 0417/11 May/June 2016 [3]

27 A customer wishes to check the balance of their bank account and inserts their card into the ATM.

(a) Describe the computer processing which takes place at the ATM.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

(b) Describe the computer processing which takes place at the bank's computer.

.....

.....

.....

.....

.....

.....

[3]

Q 8 - Page 5 0417/12 May/June 2016

28

When a bank cheque is presented to a bank it is sent off to a cheque clearing centre.

(a) (i) Give **three** items of information that are pre-printed on the cheque in magnetic ink.

1

.....

2

.....

3

.....

[3]

(ii) Describe how MICR is used to read these details from the cheque.

.....

.....

.....

.....

.....

[3]

(b) Explain why magnetic ink is used rather than bar codes on a cheque.

.....

.....

.....

.....

[2]

Q 10 - Page 6 0417/12 May/June 2016

29

Identify **four** uses of online booking systems.

1

.....

2

.....

3

.....

4

.....

[4]

Q 5 - Page 3 0417/13 May/June 2016

30

A school library uses a computer system where there is a book file and a borrower file.

Each student has a unique membership number and each book has a unique accession number. The accession number is the number given to a book when it is first bought by the library.

When a book is borrowed, the student has their thumbprint read by a special input device.

(a) Describe the computer processing involved when a book is borrowed.

.....

.....

.....

.....

.....

.....

[3]

(b) Describe the computer processing involved when a book is returned. There is no need for the borrower to have their thumbprint read during this process.

.....

.....

.....

.....

[2]

Q 8 - Page 5 0417/13 May/June 2016

31

When customers use phone banking they are connected to an automated system.

(a) Give **two** items of data they are asked to key in to identify themselves to the system.

1

.....

2

.....

[2]

(b) Identify **three** services that they can use in phone banking.

1

.....

2

.....

3

.....

[3]

Q 9 - Page 6 0417/13 May/June 2016

32

A student exam answer sheet consists of boxes or lozenges which the student shades in to indicate their choice of answer.

- (a) Apart from the details of the test and the lozenges, give **two** items of information, essential to the student, that are pre-printed on the answer sheet.

1

.....

2

.....

[2]

- (b) Describe how Optical Mark Recognition is used to process the details from the form.

.....

.....

.....

.....

.....

.....

[3]

- (c) Describe the drawbacks of using Optical Mark Recognition.

.....

.....

.....

.....

.....

.....

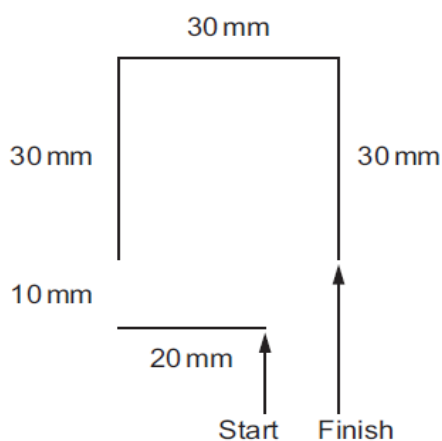
[3]

Q 10 - Page 7 0417/13 May/June 2016

33

A floor turtle can use the following instructions:

INSTRUCTION	MEANING
FORWARD n	Move n mm forward
BACKWARD n	Move n mm backward
LEFT t	Turn left t degrees
RIGHT t	Turn right t degrees
PENUP	Lift the pen
PENDOWN	Lower the pen
REPEAT n	Repeat the following instructions n times
END REPEAT	Finish the REPEAT loop



Complete the set of instructions to draw this shape by filling in the blank lines. Do **not** use additional lines.

PENDOWN
 LEFT 90
 FORWARD 20
 RIGHT 90
 PENUP
 FORWARD 30

[6]

Q 6 - Page 4 0417/11 May/June 2015

34

A home is fitted with a microprocessor-controlled burglar alarm system. It is not connected to a police station.

(a) Tick **three** sensors which would be used in such a system.

	☒
Pressure sensor	☐
Oxygen level sensor	☐
Wind speed sensor	☐
Sound sensor	☐
Body sensor	☐
Moisture sensor	☐
Infra-red sensor	☐
Touch sensor	☐

[3]

(b) Describe how a microprocessor-controlled burglar alarm system operates, referring to those sensors you identified in part (a).

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[5]

Q 7 - Page 5 0417/11

May/June 2015

35 Tick **four** drawbacks to car manufacturers of introducing robots on production lines.

	✓
Redundancy payments to previous workers are expensive	
The cars produced are not of a consistent standard	
Robots are unable to think for themselves	
Robots do not go on strike	
Robots are expensive to buy	
Mistakes are never made	
Maintaining robots costs money	
Car workers have to be paid more	

[4]

Q 10 - Page 9 0417/11

May/June 2015

36

A lending library has introduced an e-book borrowing facility. Borrowers will be able to download a book to their tablet computer or e-book reader. It will automatically remove itself from the device after the loan period has expired.

Discuss the advantages and disadvantages of such a system to the library and the borrower.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

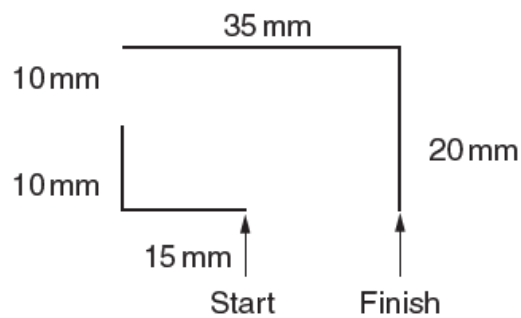
[6]

Q 14 - Page 13 0417/11 May/June 2015

37

A floor turtle can use the following instructions:

INSTRUCTION	MEANING
FORWARD n	Move n mm forward
BACKWARD n	Move n mm backward
LEFT t	Turn left t degrees
RIGHT t	Turn right t degrees
PENUP	Lift the pen
PENDOWN	Lower the pen



The turtle is facing up the page and the pen is up.

A student has written the following set of instructions to make the floor turtle draw the above shape.

1. PENDOWN
2. LEFT 90
3. FORWARD 15
4. RIGHT 90
5. PENUP
6. FORWARD 20
7. RIGHT 90
8. PENUP
9. FORWARD 90
10. RIGHT 35
11. FORWARD 20

Although the first **four** instructions are correct, this set of instructions would not draw the shape exactly as shown. **One** instruction has been missed out and **four** are incorrect.

(a) Write down the missing instruction and indicate where it should be positioned in the list of instructions provided.

.....
 [1]

(b) Identify the **four** incorrect instructions and state the correct instructions to be used.

1

 2

 3

 4
 [4]

Q 6 - Page 4 0417/12 May/June 2015

A microprocessor controlled washing machine is fitted with a number of sensors.

(a) Tick **three** sensors which would be used in such a system.

	✓
Pressure sensor	
Height sensor	
Wind speed sensor	
Sound sensor	
Temperature sensor	
Moisture sensor	
Cold sensor	
Detergent sensor	

[3]

(b) Describe the actions taken by a microprocessor during a wash cycle, based on the readings from those sensors identified in part **(a)**.

.....[5]

Q 7 - Page 8 0417/12
May/June 2015

A bank has introduced a system whereby its customers will use contactless smart cards to make purchases. This method of payment involves customers holding their card close to a reader at shopping outlets and restaurants.

.....[6]

Q 8 - Page 7 0417/12
May/June 2015

An oil prospecting company wants to use an expert system to help it to decide where to drill for oil.

[6]

1
2 [2]

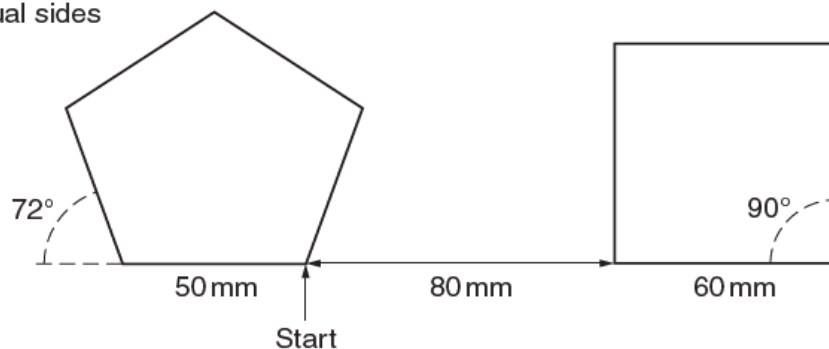
May/June 2015

40

A floor turtle can use the following instructions:

INSTRUCTION	MEANING
FORWARD n	Move n mm forward
BACKWARD n	Move n mm backward
LEFT t	Turn left t degrees
RIGHT t	Turn right t degrees
PENUP	Lift the pen
PENDOWN	Lower the pen
REPEAT n	Repeat the following instructions n times
END REPEAT	Finish the REPEAT loop

Each shape
has equal sides



The turtle is facing up the page and the pen is up.

A student has written the following set of instructions to make the floor turtle draw the above shapes.

1. PENDOWN
2. LEFT 90
3. REPEAT 6
4. FORWARD 50
5. RIGHT 72
6. BACKWARD 80
7. RIGHT 90
8. FORWARD 60
9. END REPEAT

Although the first **two** instructions are correct, this set of instructions would not draw the shapes exactly as shown. **Four** instructions have been missed out and **one** is incorrect.

(a) Identify the incorrect instruction and state the correct instruction to be used.

.....
[1]

(b) Write down the **four** missing instructions and indicate where they should be positioned in the list of instructions provided.

1

 2

 3

 4

[4]

Q 8 - Page 6 0417/13
 May/June 2015

41

An office has a microprocessor controlled central heating system.

(a) Name and describe, in detail, the use of **two** input devices in such a system.

Name 1

.....

Use 1

.....

Name 2

.....

Use 2

.....

[4]

Describe how the microprocessor would keep the temperature of the office at a constant 19°C.

.....

.....

.....

.....

.....

.....

.....

..... [4]

Q 9 - Page 7 0417/13 May/June 2015

A travel company uses an online booking system to book flights for its customers.

- Describe the processing and outputs involved in such a system.

[6]

- 1
2
3 [3]

Q 8 - Page 6 0417/13
May/June 2015

- 43 A headteacher has decided that the school newsletter will no longer be printed out for parents but, instead, will appear on the school website.
- Name **three** features available on a website not available in a hard copy version of the newsletter. For each one describe how it could be used on the school website.
- Feature 1
- Use
-
- Feature 2
- Use
-
- Feature 3
- Use
-
- [6]

Q 17 - Page 15 0417/13
May/June 2015

44

A geyser is a hot spring in which water occasionally boils, sending a tall column of water and steam into the air. A temperature sensor is used to monitor the temperature near the geyser. The data from the sensor is sent to a microprocessor.

(a) Explain why the data from the sensor has to be changed before it is read by the microprocessor.

.....

.....

.....

.....

.....

.....

[3]

(b) Describe five advantages of using sensors and microprocessors to monitor the temperature in geysers rather than using manual methods.

1

.....

2

.....

3

.....

4

.....

5

.....

[5]

Q 5 - Page 3 0417/11 May/June 2018

44

A fish farm has a computer system that constantly monitors the quality of the water in its fish ponds. The system uses sensors to carry out the monitoring.

(a) Name **two** sensors that could be used to monitor the quality of the water in the fish ponds.

1

2

[2]

(b) Name the device that is used to convert the data from the sensor into a form that could be read by the computer.

..... [1]

(c) Describe the role of the computer in monitoring the quality of the water in the fish ponds.

.....

.....

.....

.....

.....

..... [3]

Q 5 - Page 4 0417/13 May/June 2018

45

Adinda is a marathon runner. She uses a smartwatch which is a smaller version of her smartphone. She trains by running in areas near her home. She uses a smartwatch with an in-built GPS system to navigate the route she is running.

(a) Describe the disadvantages of using a smartwatch in this way.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

(b) Name two other uses of GPS systems.

1.....

2.....

[2]

(c) Adinda also carries a smartphone strapped to her arm but finds this is difficult to use when running. A smartphone can be used for communication, either for making phone calls or text messaging. Adinda stops part way through her run for a rest and wishes to communicate with her friend using the smartphone.

Describe why text messaging is a better way for Adinda to communicate with her friend rather than making a phone call.

.....

.....

.....

.....

.....

.....

[3]

Mr. Ayman Alnahhas